

Technical Datasheet - VBF-T3R2-DSH-01

Power-tool battery cell - NMC811 | graphite pouch (virtual design, Chen2020-based simulation). Values mechanically taken from parameter set / simulation outputs; missing items honestly marked.

Field	Value	Source
Rated capacity (Ah)	3.32 (nominal parameter); 3.328 (1C DFN measured)	cell/r2_p1_params_final.json; cell/r2_p1_1c_dfn.json:capacity_ah
Nominal voltage / voltage window (V)	window 2.5 - 4.2 V; discharge midpoint voltage 3.886 V	parameter set; cell/r2_p1_energy.json:midpoint_voltage_v
Rated energy (Wh)	12.13	cell/r2_p1_energy.json:energy_wh (time integration of V-I at 1C)
Energy density (Wh/kg)	407.6 (contract caliber: cell mass excludes electrolyte)	cell/r2_p1_energy.json:energy_density_wh_kg
Volumetric energy density (Wh/L)	837.4	cell/r2_p1_energy.json:energy_density_wh_l
Maximum continuous discharge rate	5C (16.61 A) verified: 98.89 % capacity retention, peak temperature 49.1 °C	cell/r2_p1_derived.json:capacity_retention_5c; t_max_5c_k
Fast-charge capability	4C (13.29 A) charge at 45 °C ambient, 900 s protocol: no lithium plating (anode potential min +36.4 mV), peak temperature 59.4 °C (<= 60 °C limit)	cell/r2_p1_4c_dfn.json:anode_potential_v, T_max_K
DC internal resistance	2.265 mOhm (start-OCV to 10 % -time drop over 1C current)	cell/r2_p1_energy.json:dcr_ohm
Peak power density	63,130 W/kg ($V_{OC}^2 / (4 \cdot DCR) / \text{mass}$, contract formula)	cell/r2_p1_energy.json:power_density_w_kg
Operating temperature range	25 °C discharge / 45 °C charge ambient verified by simulation; wider range not verified - quoted range is 25-45 °C (simulation conditions only)	protocol conditions of the run files
Cycle life	Not simulated (aging-capable parameter set available; not part of task contract) - must not be fabricated	honest annotation
Safety determination	4C/45 °C fast charge: not plated, T_max within 60 °C; abuse scenarios (overcharge/nail/crush/drop) not simulated for this task - N/A beyond pure simulation boundary	cell/r2_p1_4c_dfn.json
Dimensions	electrode area 0.1027 m ² (0.065 m x 1.58 m); layer stack thickness 141 um; shell/footprint Not provided	parameter set + cell/r2_p1_energy.json:thickness_m
Mass	29.75 g cell (contract caliber, electrolyte excluded; with literature-density electrolyte ~35.7 g)	cell/r2_p1_energy.json:mass_kg + BOM

Notes: this datasheet describes a virtual cell designed and verified within the pure-simulation boundary (Model: DFN, lumped thermal, irreversible plating option). Physical prototyping, cycle life, calendar aging, mechanical abuse and pack integration are outside this boundary and must be verified experimentally before production.